Time-independent Lindblad embeddings of two-parameter Werner–Holevo noise

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Abstract

An orthogonally covariant quantum channel need not have an orthogonally covariant time-independent generator. This distinction matters when deciding whether a calibrated noise channel can arise from one constant Lindblad master equation. We consider the two-parameter mixture of the identity and the symmetric and antisymmetric Werner–Holevo channels on a finite d-level system. For positive eigenvalues s and a on the symmetric traceless and antisymmetric operator sectors, respectively, we prove that an arbitrary same-system Lindblad embedding exists exactly when $s^{(d+2)/d} \leq a \leq s^{(d-2)/d}$. The covariant generator and its rate cone are established prior results; the additional step is necessity without imposing covariance, a Hamiltonian restriction, or a logarithm branch. Real sector traces fix the average logarithmic rates, and averaging over the full orthogonal group then produces the principal covariant generator. Odd sector multiplicities exclude every remaining signed qutrit case, yielding a complete classification for $d = 3$. We give normalized jump operators and illustrate the qutrit region. Negative even-multiplicity sectors in higher dimensions remain outside the classification.

Keywords: quantum channels; Lindblad embedding; Werner–Holevo noise; orthogonal covariance; quantum Markov semigroups

1 Endpoint noise and a constant master equation

A quantum channel at one finite time does not by itself specify a time-independent master equation. The endpoint embedding problem asks whether $\Phi = e^{T\mathcal{L}}$ for a Gorini–Kossakowski–Lindblad–Sudarshan (GKLS) generator $\mathcal{L}$ on the same system. General logarithm and conditional complete-positivity tests are available [1], but degeneracies allow substantial freedom in the logarithm. In particular, symmetry of an endpoint cannot simply be imposed on an unknown generator.

The noisy Werner–Holevo models provide a concrete setting for this issue [2, 3]. Their decomposition into orthogonal operator sectors is established, including in real randomized benchmarking [4]. Bhat, Chakraborty and Franz studied the generator family spanned by depolarization, transposition and the identity, and derived its semigroup exponential and positivity conditions [5]. Thus the covariant family and the associated rate cone are already known. What remains to justify for an endpoint test is that an arbitrary, possibly noncovariant GKLS logarithm cannot evade that cone.

Here we prove this reduction for the positive-spectrum part of the two-parameter model. The argument uses real traces on degenerate endpoint sectors rather than an enumeration of generic-spectrum logarithm branches. It gives a complete endpoint classification for qutrits after the signed sectors are excluded. Our result concerns a single finite, time-independent generator; it does not classify time-dependent completely positive divisible paths. The related Pauli-channel semigroup criteria of Kattemölle, Gulácsi and Burkard (arXiv:2602.08464v1) concern a different operator algebra and supply a useful conceptual comparison.

2 Model, sectors and physical domain

Fix an orthonormal computational basis of $\mathbb{C}^d$, with $d \geq 3$. Transposition below is the complex-linear map $X \mapsto X^{\mathsf{T}}$ in this basis, not Hermitian conjugation. Following Ref. [3], define

$$W_-(X) = \frac{\mathrm{Tr}(X) I_d - X^{\mathsf{T}}}{d-1}, \qquad W_+(X) = \frac{\mathrm{Tr}(X) I_d + X^{\mathsf{T}}}{d+1}, \tag{1}$$

$$\Phi_{d,x,y} = (1-x-y)\,\mathrm{id} + xW_- + yW_+, \qquad x>0, \quad y>0, \quad x+y \leq 1. \tag{2}$$

All inputs in $M_d(\mathbb{C})$ are included. In particular, complete positivity is required on entangled inputs. No enlarged system, postselection or approximation is allowed in the embedding.

Set

$$q = 1-x-y, \qquad c = \frac{x}{d-1} + \frac{y}{d+1}, \qquad b = \frac{y}{d+1} - \frac{x}{d-1}. \tag{3}$$

Then $\Phi(X) = qX + c\,\mathrm{Tr}(X) I_d + bX^{\mathsf{T}}$. The orthogonal superoperator projectors are

$$\mathsf{P}_0(X) = \frac{\mathrm{Tr}(X)}{d} I_d, \qquad \mathsf{P}_{\mathrm{s}}(X) = \frac{X + X^{\mathsf{T}}}{2} - \mathsf{P}_0(X), \qquad \mathsf{P}_{\mathrm{a}}(X) = \frac{X - X^{\mathsf{T}}}{2}. \tag{4}$$

Their complex dimensions, also their dimensions on real Hermitian bases, are

$$1, \qquad m_{\mathrm{s}} = \frac{d(d+1)}{2} - 1, \qquad m_{\mathrm{a}} = \frac{d(d-1)}{2}. \tag{5}$$

The traceless Hermitian bases may be chosen as real symmetric matrices and purely imaginary antisymmetric matrices. The channel is

$$\Phi = \mathsf{P}_0 + s\mathsf{P}_{\mathrm{s}} + a\mathsf{P}_{\mathrm{a}}, \tag{6}$$

$$s = q + b = 1 - \frac{d}{d-1}x - \frac{d}{d+1}y,$$

$$a = q - b = 1 - \frac{d-2}{d-1}x - \frac{d+2}{d+1}y. \tag{7}$$

Both s and a are strictly less than one in the domain (2); their signs are not fixed by complete positivity.

For clarity, use the unnormalized Choi convention

$$J(\mathcal{M}) = \sum_{i,j=1}^{d} E_{ij} \otimes \mathcal{M}(E_{ij}), \qquad |\Omega\rangle = \sum_{i=1}^{d} |ii\rangle, \tag{8}$$

with input followed by output. If F is the swap, then $J(\Phi) = q|\Omega\rangle\langle\Omega| + cI + bF$. Its eigenvalues are

$$dq + \frac{2y}{d+1}, \qquad \frac{2y}{d+1}, \qquad \frac{2x}{d-1}, \tag{9}$$

with multiplicities $1, m_{\mathrm{s}}, m_{\mathrm{a}}$. They are strictly positive even on $x+y=1$, and $\mathrm{Tr}_{\mathrm{out}}\, J(\Phi) = I_d$. A full-rank Choi matrix therefore does not imply an invertible channel superoperator or a Lindblad embedding.

We set $T = 1$ and $\hbar = 1$. A generator has the form

$$\mathcal{L}(X) = -i[H, X] + \sum_r \left(V_r X V_r^\dagger - \frac{1}{2}\{V_r^\dagger V_r, X\} \right), \qquad H = H^\dagger. \tag{10}$$

For a general observation time $T > 0$, divide the generator and its jump rates below by T, and divide the displayed jumps by $\sqrt{T}$. The endpoint weights and eigenvalues are dimensionless.

3 Unrestricted positive-spectrum criterion

Theorem 1. *For every integer $d \geq 3$, let $\Phi_{d,x,y}$ satisfy (2) and suppose $s > 0$ and $a > 0$. There exists a finite GKLS generator $\mathcal{L}$ on $M_d(\mathbb{C})$ with $e^{\mathcal{L}} = \Phi_{d,x,y}$ if and only if*

$$s^{(d+2)/d} \leq a \leq s^{(d-2)/d}. \tag{11}$$

Necessity allows arbitrary Hamiltonians, jump operators, logarithm branches and transformations within degenerate endpoint eigenspaces. A generator can be chosen with $H = 0$ and $\mathcal{L}_0 = (\log s)\mathsf{P}_{\mathrm{s}} + (\log a)\mathsf{P}_{\mathrm{a}}$.

The cone in (11) is the endpoint expression of the known covariant-generator cone [5]. The content of the theorem is that the same cone is necessary for every generator of this endpoint.

3.1 Real sector traces and the orthogonal average

First assume $s \neq a$. Since any matrix commutes with its exponential, $\mathcal{L}$ commutes with Φ. The distinct eigenvalues $1, s, a$ force $\mathcal{L}$ to preserve the three sectors in (4). On the scalar sector $\mathcal{L}(I_d) = hI_d$; trace preservation gives $h = 0$. Hermiticity preservation makes the restrictions $\mathcal{L}_{\mathrm{s}}$ and $\mathcal{L}_{\mathrm{a}}$ real matrices in their Hermitian bases. Consequently their traces are real and

$$\begin{aligned} e^{\operatorname{Tr}\mathcal{L}_{\mathrm{s}}} &= \det e^{\mathcal{L}_{\mathrm{s}}} = s^{m_{\mathrm{s}}}, & \operatorname{Tr}\mathcal{L}_{\mathrm{s}} &= m_{\mathrm{s}} \log s, \\ e^{\operatorname{Tr}\mathcal{L}_{\mathrm{a}}} &= \det e^{\mathcal{L}_{\mathrm{a}}} = a^{m_{\mathrm{a}}}, & \operatorname{Tr}\mathcal{L}_{\mathrm{a}} &= m_{\mathrm{a}} \log a. \end{aligned} \tag{12}$$

These identities do not select a logarithm branch or assume normality of the restrictions.

The endpoint has full orthogonal covariance: $\mathrm{Ad}_O(X) = OXO^{\mathsf{T}}$ commutes with Φ for every $O \in \mathrm{O}(d)$. Average the generator, with normalized Haar measure,

$$\overline{\mathcal{L}} = \int_{\mathrm{O}(d)} \mathrm{Ad}_O^{-1}\, \mathcal{L}\, \mathrm{Ad}_O \, dO. \tag{13}$$

Unitary conjugation preserves GKLS form. The finite-dimensional GKLS cone is closed and convex: trace and Hermiticity preservation are linear conditions and conditional complete positivity is a linear matrix inequality [1, 5]. Thus $\overline{\mathcal{L}}$ is again GKLS. It preserves the sectors, vanishes on the scalar sector, and retains their restricted traces.

We spell out why its restrictions are scalar for the full group. Diagonal sign matrices act on each off-diagonal pair (i, j) with character $\varepsilon_i\varepsilon_j$. These characters are distinct for the full set of independent signs, so an intertwiner within either sector scales each off-diagonal basis vector. Permutation matrices make those scales equal. In the symmetric sector, the diagonal traceless subspace is the sum-zero permutation representation; an intertwiner is scalar there as well. A planar rotation by $\pi/4$ maps $E_{ii} - E_{jj}$ to a signed multiple of $E_{ij} + E_{ji}$, equating the diagonal and off-diagonal symmetric scales. The antisymmetric sector has no diagonal component. This elementary argument gives a scalar on each sector for all $d \geq 3$. In $d = 4$, using full $\mathrm{O}(4)$ is essential: reflections exchange the two Hodge components that are separately invariant under $\mathrm{SO}(4)$. The corresponding orthogonal-sector decomposition is also described in Ref. [4].

Equation (12) now fixes the two scalar restrictions:

$$\overline{\mathcal{L}} = (\log s)\mathsf{P}_{\mathrm{s}} + (\log a)\mathsf{P}_{\mathrm{a}} = \mathcal{L}_0. \tag{14}$$

Therefore an arbitrary GKLS embedding implies that the principal sector logarithm is GKLS. This conclusion does not follow from a general commutation of averaging and exponentiation; (14) follows from the specific real trace identities.

When $s = a > 0$, the distinct-sector argument is unnecessary. The channel is depolarizing and $\mathcal{L}_0 = (\log s)(\mathrm{id} - \mathsf{P}_0)$ is an explicit GKLS generator. Because $0 < s < 1$, (11) holds automatically. This handles the coincident traceless eigenvalues without assuming sector preservation for an arbitrary logarithm on that line.

3.2 The known rate cone in endpoint variables

Let $\ell_{\mathrm{s}} = \log s$, $\ell_{\mathrm{a}} = \log a$ and $Q = I - |\Omega\rangle\langle\Omega|/d$. Direct Choi reshuffling yields

$$\begin{aligned} J(\mathcal{L}_0) &= \frac{\ell_{\mathrm{s}} + \ell_{\mathrm{a}}}{2}|\Omega\rangle\langle\Omega| + \frac{\ell_{\mathrm{s}} - \ell_{\mathrm{a}}}{2}F - \frac{\ell_{\mathrm{s}}}{d}I, \\ QJ(\mathcal{L}_0)Q &= \gamma_{\mathrm{s}}C_{\mathrm{s}} + \gamma_{\mathrm{a}}C_{\mathrm{a}}, \end{aligned} \tag{15}$$

$$\begin{aligned} C_{\mathrm{s}} &= \frac{I+F}{2} - \frac{|\Omega\rangle\langle\Omega|}{d}, & C_{\mathrm{a}} &= \frac{I-F}{2}, \\ \gamma_{\mathrm{s}} &= \frac{(d-2)\ell_{\mathrm{s}} - d\ell_{\mathrm{a}}}{2d}, & \gamma_{\mathrm{a}} &= \frac{d\ell_{\mathrm{a}} - (d+2)\ell_{\mathrm{s}}}{2d}. \end{aligned} \tag{16}$$

The Choi-sector projectors $C_{\mathrm{s}}, C_{\mathrm{a}}$ are mutually orthogonal and distinct from the superoperator notation $\mathsf{P}_{\mathrm{s}}, \mathsf{P}_{\mathrm{a}}$. Conditional complete positivity is exactly $\gamma_{\mathrm{s}}, \gamma_{\mathrm{a}} \geq 0$, which is equivalent to (11). This proves necessity for $s \neq a$.

To make the prior-result correspondence explicit, write $\mathcal{T}(X) = X^{\mathsf{T}}$. Section 6.3 of Ref. [5] uses

$$\mathcal{G}_{\alpha,\eta} = \alpha\mathsf{P}_0 + \eta\mathcal{T} - (\alpha+\eta)\,\mathrm{id}\,. \tag{17}$$

The exact parameter map is

$$\alpha = -\ell_{\mathrm{s}}, \qquad \eta = \frac{\ell_{\mathrm{s}} - \ell_{\mathrm{a}}}{2}, \qquad \gamma_{\mathrm{s}} = \frac{\alpha}{d} + \eta, \qquad \gamma_{\mathrm{a}} = \frac{\alpha}{d} - \eta. \tag{18}$$

Thus its covariant generator cone $\alpha \geq d|\eta|$ is precisely (11). The symbol η denotes the parameter called γ in that reference, to avoid confusing it with the two jump rates here. Its exponential has sector eigenvalues $e^{-\alpha t}$ and $e^{-(\alpha+2\eta)t}$. Neither this exponential nor its covariant rate cone is claimed as a new result.

3.3 Normalized jumps and sufficiency

Choose Frobenius-orthonormal Hermitian bases $\{S_\mu\}_{\mu=1}^{m_{\mathrm{s}}}$ of real symmetric traceless matrices and $\{A_\nu\}_{\nu=1}^{m_{\mathrm{a}}}$ of purely imaginary antisymmetric matrices. Their completeness identities are

$$\begin{aligned} \sum_\mu S_\mu X S_\mu^\dagger &= \frac{\mathrm{Tr}(X)I_d + X^{\mathsf{T}}}{2} - \frac{X}{d}, & \sum_\mu S_\mu^\dagger S_\mu &= \frac{m_{\mathrm{s}}}{d}I_d, \\ \sum_\nu A_\nu X A_\nu^\dagger &= \frac{\mathrm{Tr}(X)I_d - X^{\mathsf{T}}}{2}, & \sum_\nu A_\nu^\dagger A_\nu &= \frac{m_{\mathrm{a}}}{d}I_d. \end{aligned} \tag{19}$$

For example, off-diagonal basis elements are $(E_{ij} + E_{ji})/\sqrt{2}$ and $i(E_{ij} - E_{ji})/\sqrt{2}$. A diagonal symmetric basis is

$$S_k^{\mathrm{diag}} = \frac{\sum_{i=1}^k E_{ii} - kE_{k+1,k+1}}{\sqrt{k(k+1)}}, \qquad 1 \leq k \leq d-1. \tag{20}$$

If (11) holds, take $H = 0$ and jumps

$$V_\mu = \sqrt{\gamma_{\mathrm{s}}}\,S_\mu, \qquad V_\nu = \sqrt{\gamma_{\mathrm{a}}}\,A_\nu. \tag{21}$$

Substituting (19) into (10) gives exactly $\mathcal{L}_0 = \ell_{\mathrm{s}}\mathsf{P}_{\mathrm{s}} + \ell_{\mathrm{a}}\mathsf{P}_{\mathrm{a}}$. Orthogonality of the superoperator projectors gives

$$e^{t\mathcal{L}_0} = \mathsf{P}_0 + s^t\mathsf{P}_{\mathrm{s}} + a^t\mathsf{P}_{\mathrm{a}}, \qquad e^{\mathcal{L}_0} = \Phi. \tag{22}$$

This proves sufficiency and completes Theorem 1. On the two boundaries of the cone, one of the jump families has zero rate; a finite generator still exists.

4 Complete qutrit classification

Corollary 2. *For $d = 3$ throughout $x > 0, y > 0, x + y \leq 1$, a finite time-independent same-qutrit GKLS embedding exists exactly when*

$$s = 1 - \frac{3}{2}x - \frac{3}{4}y > 0, \qquad a = 1 - \frac{1}{2}x - \frac{5}{4}y > 0, \qquad s^5 \leq a^3 \leq s. \tag{23}$$

Proof. The two real Hermitian sector multiplicities are $m_{\mathrm{s}} = 5$ and $m_{\mathrm{a}} = 3$. A zero eigenvalue makes Φ singular, which is impossible for a finite matrix exponential. If one eigenvalue is negative, then $s + a = 2q \geq 0$ prevents both from being negative; the nonzero eigenvalues are therefore distinct and the sector restrictions of an embedding would be real. A real matrix exponential has positive determinant. A negative scalar on either odd-dimensional sector has negative determinant, so no such logarithm exists. All remaining possibilities have $s, a > 0$ and Theorem 1 gives (23). □

Figure 1 separates full Choi rank from homogeneous embeddability. At $x = y = 1/4$, the endpoint parameters are $s = 7/16$ and $a = 9/16$, and both rate inequalities are strict. This same input reproduces the published output benchmark in Ref. [3]: the pure state $|\psi\rangle = (i, 1, 0)^{\mathsf{T}}/\sqrt{2}$ has output spectrum $\{1/8, 3/16, 11/16\}$. This is a normalization check, rather than a new capacity result.

In contrast, the legal point $x = 1/2, y = 1/10$ gives $s = 7/40$ and $a = 5/8$, with $a^3 = 125/512 > s$. Its principal symmetric jump rate is negative. Theorem 1 shows that a Hamiltonian, noncovariant jumps or a different logarithm branch cannot repair that failure. The statement concerns the exact channel at a specified time; applying it to a noisy experimental estimate would require uncertainty propagation and an independent model-validity assessment.

5 Interpretation and limitations

The result converts a known covariant-generator cone into an unrestricted endpoint test for the positive orthogonal sectors. This distinction is physically useful when the measured or prescribed endpoint has symmetry but the underlying master equation is not assumed to share it. The construction uses a zero Hamiltonian, while the exclusion statement allows any Hamiltonian and any finite set of jumps on the same d-level system. It is exact in finite dimension, with no perturbative, continuum or long-time approximation.

The arbitrary-generator reduction uses positivity of s and a. For an even-dimensional negative sector, a real logarithm may have trace $m \log |\lambda|$. An orthogonal average would then exponentiate to the absolute-spectrum endpoint, not necessarily to the original signed channel. Thus the present proof does not decide negative even-multiplicity sectors for $d \geq 4$. The full signed classification in arbitrary dimension remains unresolved. The criterion applies to a finite generator on the original system and does not characterize time-dependent divisible paths.

The orthogonal sectors are established in Ref. [4], the covariant cone and exponential in Ref. [5], and the general conditional-positivity test in Ref. [1]. The additional argument is the branch-independent real-trace reduction in (12)–(14) and its qutrit consequence. Priority is limited to the compared literature; no claim of a comprehensive classification of symmetric quantum-channel embeddings is made.

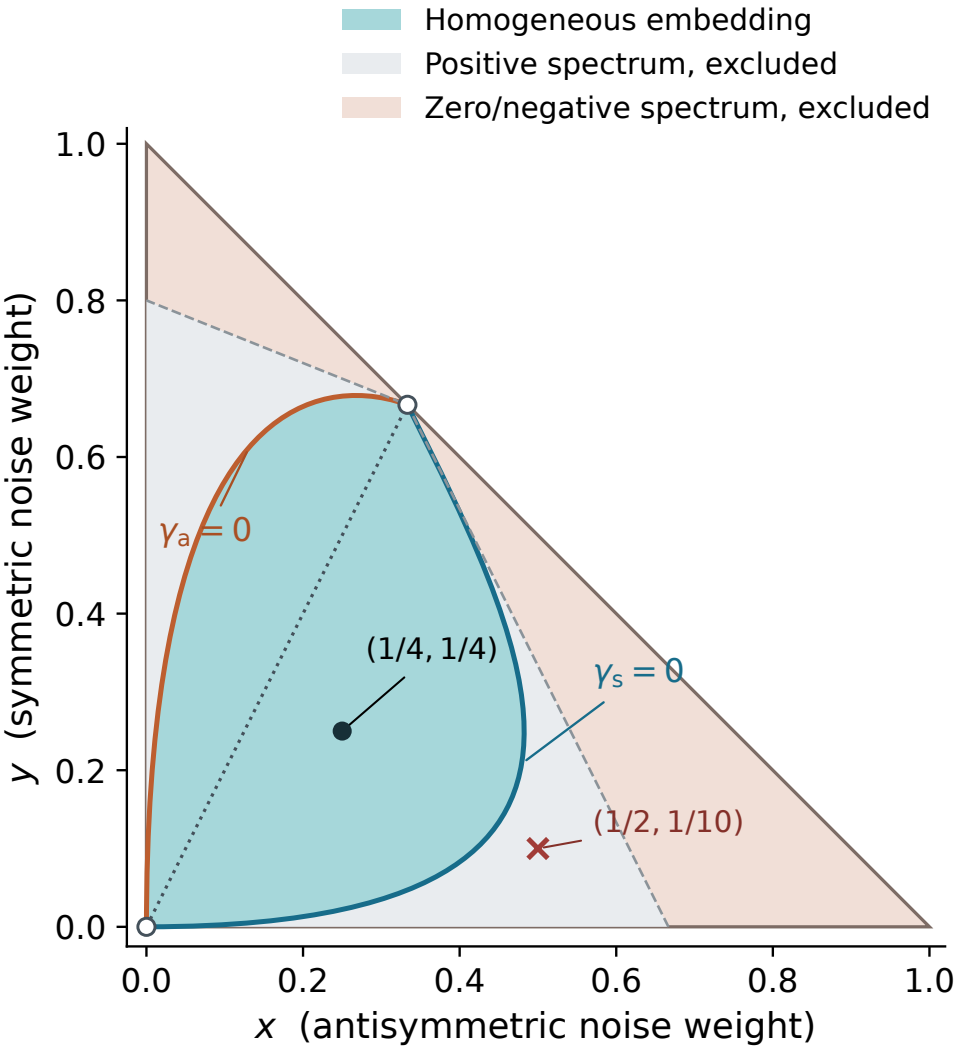


Figure 1: Exact homogeneous-embedding region in the qutrit noise triangle. The colored region includes its two rate boundaries wherever $x, y > 0$; the axes and the limiting point $(1/3, 2/3)$ are excluded. The boundary $\gamma_s = 0$ has $a^3 = s$, and $\gamma_a = 0$ has $a^3 = s^5$. The dotted line $y = 2x$ is the depolarizing subfamily. The circle at $(1/4, 1/4)$ is embeddable, whereas the cross at $(1/2, 1/10)$ has positive spectrum but violates the cone. Curves are sampled from the analytic boundary equations for rendering; the classification follows from Corollary 2.

Data, code and generative-AI disclosure

The proof is analytic. The accompanying source package contains exact matrix-unit and Kraus reconstructions, symbolic Choi and jump checks, and code and rational coordinate data for Fig. 1. Its reproducibility instructions specify the environments and distinguish exact identities from optional floating-point diagnostics. No experimental dataset is used.

OpenAI Codex was used in literature searching and comparison, formulation of the reduction, derivations, symbolic and numerical code, checking, and manuscript drafting. These uses extend beyond copy editing. Separate AI review passes checked the scientific argument, but do not constitute external peer review. Final scientific review, authorship confirmation and accountability by the named human author remain pending for this local draft.

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